

SOFTWARE REQUIREMENTS SPECIFICATION

COLLABORATIVE KNOWLEDGE ECOSYSTEM

Study Buddy

A modern academic "Second Brain" designed for continuous learning, peer-to-peer resource optimization, and centralized intellectual capital for students.

PREPARED FOR

Software Engineering Internship Evaluation

1. INTRODUCTION

In contemporary higher education, the management of extensive study repositories is critical for academic excellence. This Software Requirements Specification (SRS) establishes a baseline for the development of **Study Buddy**, an advanced web application designed as a student's personalized and collaborative "Second Brain."

1.1 Purpose

This document details the functional, non-functional, operational, and architectural requirements for Study Buddy. It serves as a definitive roadmap for the engineering lifecycle, aligning implementation paradigms with user expectations.

1.2 Scope

Study Buddy is a centralized, access-controlled ecosystem tailored for higher-education environments. It provides seamless management of course materials, multi-format note parsing, full-text indexed discovery, and peer-to-peer review systems. It functions as a modular platform, integrating standard client web architectures with a resilient persistent database layer.

1.3 Intended Users

The primary audiences for this specification include undergraduate and postgraduate students seeking optimized academic storage, student contributors archiving reference materials, software developers establishing backend data pipelines, and internal administrators supervising data integrity.

2. PROBLEM STATEMENT

Core Challenge: Academic information fragmentation across decentralized channels compromises retention, reduces productivity, and leads to duplicated preparation effort.

Students rely on ad-hoc networks like WhatsApp groups, transient Telegram channels, disorganized cloud directories, and local drives to store material. This creates information silos where valuable notes shared by seniors or peers are lost over semesters. Without unified semantic indexing or categorization, students waste hours hunting down authentic references, causing inconsistent exam preparation and disjointed peer collaborative learning.

3. PROJECT OBJECTIVES

The primary objectives of Study Buddy are established to directly address material fragmentation through the following key mandates:

- **Centralization:** Construct an instantly retrievable master repository consolidating academic content (PDFs, presentations, structured images).

- **Collaborative Knowledge Distribution:** Enable peer crowdsourcing through upload, download, bookmarking, and native feedback loops.
- **Quality Curation:** Implement dual-layered validation combining public community ratings with admin-approved curation workflows.
- **Academic Synergy:** Establish a structured framework categorizing materials precisely by subject matrices, curricula tags, and course semesters.

4. OVERALL SYSTEM DESCRIPTION

4.1 Product Perspective

Study Budy operates as an independent, web-based software application featuring a decoupled three-tier architecture: the User Presentation Interface, the Application Logic Controller layer, and the Relational Database Storage environment. It acts as an extensible framework designed to support seamless upgrades, such as external cloud connectors and analytical models.

4.2 Product Features

The platform's high-level features include user identity provisioning, interactive dashboard rendering, file upload parsing engines, multi-variable classification filters, structural social interaction modules (comments, ratings, saves), and a secure administrative dashboard interface.

4.3 User Classes

- **Standard Student / Peer User:** Authenticated end-users authorized to discover material, execute downloads, contribute personal archives, and interact with notes through reactions and reviews.
- **System Administrator:** Privileged operators overseeing operational security, validating community uploads for policy compliance, and managing user access control lists.

4.4 Operating Environment

The software is cross-platform and optimized for standard web environments:

- **Client Execution:** Standard desktop/mobile browsers (Chrome v110+, Safari v16+, Firefox v112+).
- **Hosting Server environment:** Ubuntu Linux 22.04 LTS or equivalent Unix servers running Apache/Nginx web servers.
- **Database Engine:** MySQL v8.0 or equivalent relational engine supporting transactional integrity.

5. FUNCTIONAL REQUIREMENTS

The mandatory functional parameters required for the Study Budy web platform are structured in the comprehensive matrix below:

REQ ID	MODULE NAME	FUNCTIONAL SPECIFICATION DETAIL
FR-AUTH-01	Registration & Login	Allows users to create accounts via academic email and log in via secure hashed credentials using session tokens.
FR-PROF-02	Profile Management	Enables users to manage avatars, institutional details, track their uploads, and monitor saved bookmarks.
FR-UPLD-03	Multi-Format Upload	Supports secure upload of PDF, PPT, and image files up to 25MB, with metadata extraction (title, subject, tags).
FR-SRCH-04	Search Notes	Provides a search engine querying title, description, and tags with instantly populated matches.
FR-FLTR-05	Taxonomy Filters	Allows narrowing down search results dynamically by Subject, Semester, and custom user-generated tags.
FR-VIEW-06	View & Download	Integrates an in-browser PDF viewer and lets users download original assets safely.
FR-INT-07	Social Interactions	Supports interactive features including Likes, Bookmarks, and Comments to drive community engagement.
FR-RTG-08	Rating System	Enables structured user reviews based on a 1-5 scale, generating a rolling average score per note document.
FR-TRND-09	Trending Notes	Computes dynamic scores prioritizing notes with high download velocities and ratings for dashboard visibility.
FR-ADM-10	Admin Approval	Enforces an admin approval queue where uploaded files remain hidden until explicitly approved by moderators.
FR-ADM-11	User Management	Grants administrators permissions to block accounts, delete reports, and audit system logs.

6. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements specify constraints and quality attributes that govern system operation:

- **Performance:** The application must resolve text search queries within 500ms under a load of 200 concurrent active requests. Page load times should not exceed 2 seconds for cached assets.
- **Security:** All user passwords must be transformed using bcrypt hashing protocols. Data transit must be encrypted over HTTPS protocols via standard TLS 1.3 wrappers.
- **Reliability:** The platform should target an operational uptime of 99.9%, minimizing unplanned service interruptions through redundant daily database snapshots.
- **Scalability:** The underlying data layer must support horizontal partitioning, comfortably scaling up to 50,000 document index pointers without noticeable operational latency.
- **Usability:** The frontend user interface must achieve high clarity, requiring no special training. Navigation must follow standard breadcrumb schemas.
- **Availability:** Core access windows must remain open 24/7/365, reserving maintenance down-times for off-peak student windows (02:00 - 04:00 AM UTC).
- **Responsive Design:** The presentation layout must dynamically adapt across standard viewports, rendering cleanly on mobile viewports, tablets, and wide-screen desktops.

7. SOFTWARE AND HARDWARE REQUIREMENTS

The environment requirements for the local development workspace and structural deployment targets are outlined below:

7.1 Software Configuration Matrix

- **Frontend Stack:** Structured HTML5 markup, responsive CSS3 styling frameworks, and client-side JavaScript (ES6+ standard).
- **Backend Engine:** Python (utilizing the lightweight Flask framework) or object-oriented PHP 8.2+ for business logic controllers.
- **Data Store:** MySQL relational database server providing ACID-compliant data storage.
- **Development Tools:** Visual Studio Code (VS Code) IDE integrated with Git version control pipelines hosted on GitHub.

7.2 Hardware Baseline Configuration

- **Workstation Platform:** Modern Laptop or Desktop computer (Intel Core i3 / AMD Ryzen 3 or higher architectures).
- **Memory Capacities:** Minimum 4 GB RAM configured on the local system (8 GB recommended for simultaneous server-database testing).
- **Network Interfaces:** Stable broadband connection for asset hosting, CDN loading, and repository sync operations.

8. FUTURE ENHANCEMENTS

To ensure long-term utility, the roadmap outlines the following iterative component additions for Study Buddy:

1. **AI-Generated Note Summaries:** Integration of Large Language Models (LLMs) to automatically generate comprehensive abstracts upon new file ingestion.
2. **Optical Character Recognition (OCR):** Implementation of image-to-text parsers to enable deep, full-text searching inside handwritten uploads or raw images.
3. **Contextual AI Recommendations:** Machine learning engines analyzing student histories to recommend notes tailored to upcoming exams or weak areas.
4. **Cross-Platform Mobile App:** Native iOS and Android clients deployed via cross-platform frameworks to offer offline study capabilities.
5. **Object Cloud Storage Integration:** Migrating assets to scalable object stores like AWS S3 or Google Cloud Storage to optimize media asset delivery.
6. **Real-time Notification Pipelines:** Push notifications informing users when a tracked peer uploads material or when a thread gets a comment.
7. **Plagiarism Detection Filters:** Scanning incoming documents against historical indexing repositories to safeguard academic integrity.
8. **Dynamic Theme Switching:** A low-light, accessible Dark Mode theme to reduce eye strain during extended night study sessions.

9. CONCLUSION

The Study Buddy Software Requirements Specification outlines a modern, scalable, and intuitive approach to managing scattered academic knowledge. By consolidating study resources into a well-categorized "Second Brain," the platform mitigates information fragmentation, saves valuable prep time, and fosters an active peer-to-peer learning culture.

Designed within a lightweight, robust engineering stack, Study Buddy is fully achievable within an internship lifecycle. It builds a solid foundation for advanced feature sets, including AI-driven insights, full-text OCR discovery, and mobile access. This framework ensures the system remains a sustainable, valuable asset for students aiming for academic success.